Stuck Job Detection & Recovery

Timing Report — Retest dated 17 October 2026

Evidence window: 2026-10-17T21:12:57.006+05:30 through 2026-10-17T21:24:55.649+05:30 (IST). **Source basis:** the attached log capture (793 lines). No runtime assumptions were added beyond explicit log messages and timestamp subtraction.

13	26	26	13
distinct jobs	worker attempts	stuck-recovery checks	DLQ entry logs

Executive findings

- All 13 jobs have two worker pickup attempts. Each attempt logs a requested 35-second sleep, giving 26 recorded sleep intervals.
- The first stuck-recovery attempt is logged 10,293–14,331 ms; median 12,311 ms after the first sleep-start log. The second recovery attempt is logged 10,000–14,998 ms; median 14,977 ms after the retry sleep-start log.
- For each job, the second scheduler decision is 'Max retries exhausted' and the job enters the DLQ with failure cause MAX_TIME_EXCEEDED (13 DLQ-entry logs; 13 TTL logs).
- The final discard decision is logged 20,347–29,335 ms; median 27,302 ms after the initial worker-pickup log. In all jobs, this decision occurs before either associated 35-second sleep-end marker.
- After the DLQ path, both worker attempts later log a status-mismatch save rejection: 26 mismatch warnings and 26 successful completeProcessingScript markers. No 'Job result saved.' line appears in this capture.

Measured intervals

Interval	Observed values	Timestamp basis
First sleep start → first recovery-attempt log	10,293–14,331 ms; median 12,311 ms	Logger timestamp → logger timestamp
Retry sleep start → second recovery-attempt log	10,000–14,998 ms; median 14,977 ms	Logger timestamp → logger timestamp
Retry decision → retry worker pickup	1–74 ms; median 17 ms	Logger timestamp → logger timestamp
Initial worker pickup → max-retries discard decision	20,347–29,335 ms; median 27,302 ms	Logger timestamp → logger timestamp
Final discard decision → Stuck Scheduler discarded-at marker	7–103 ms; median 9 ms	Embedded epoch ms → embedded epoch ms
DLQ-entry log → discarded-with-TTL log	7–100 ms; median 8 ms	Logger timestamp → logger timestamp
recoverOneJob start → end marker	0–1 ms; median 1 ms	Embedded epoch ms → embedded epoch ms; millisecond resolution
Sleep-start → sleep-end markers	35,000–35,001 ms; median 35,000 ms	Embedded epoch ms → embedded epoch ms
DB-write-start → DB-write-end markers	0–1 ms; median 1 ms	Embedded epoch ms → embedded epoch ms; not a SQL runtime

Key precision rule. Logger timestamps are subtracted from logger timestamps; embedded epoch-millisecond values are subtracted from embedded epoch-millisecond values. The two sources are preserved separately in the CSV and are never silently reconciled.

Per-job comparison

Each duration below is calculated from the stated pair of logger timestamps. The separate CSV preserves the exact source times, embedded epochs, and source line references for each worker attempt.

Job ID (short form; full ID in CSV)	Attempt 1 sleep → check 1 (ms)	Retry decision → pickup (ms)	Attempt 2 sleep → check 2 (ms)	Pickup 1 → final discard (ms)	DLQ entry → TTL log (ms)
01a14a87…e6afb4a	13239	28	14973	28244	100
01a14a88…8a3962c	12363	1	10002	22367	12
01a14a89…c819596	11333	74	14928	26337	7
01a14a8a…5cecbc3	10345	1	10000	20347	7
01a14a8b…5e9fd48	14331	24	14977	29335	7
01a14a8c…d30d51e	13296	4	14998	28300	8
01a14a8c…a72050e	12311	65	14936	27314	7
01a14a8d…9b31aa4	11308	15	14986	26311	10
01a14a8e…f4cafd5	10293	19	14982	25295	7
01a14a8f…e8a7b14	14317	16	14986	29319	7
01a14a90…60417f5	13272	14	14987	28276	9
01a14a91…f786d1c	12298	17	14984	27302	8
01a14a92…adc21e9	11252	25	14976	26255	9

What the retest logs directly show

- Each job logs two worker pickups and two 35-second sleep-start markers. The embedded sleep-start/end epoch values differ by 35,000 ms for 25 attempts and 35,001 ms for one attempt.
- The first scheduler check occurs 10.293–14.331 seconds after the first sleep-start log. After retry pickup, the second check occurs 10.000–14.998 seconds after the retry sleep-start log.
- The retry pickup follows the retry-decision log by 1–74 ms. The second recovery path is logged as max retries exhausted, then the JobDiscardService/DlqService path logs MAX_TIME_EXCEEDED.
- The two worker attempts continue to their respective sleep-end markers after the discard/DLQ sequence, then each logs a DB-write attempt whose ResultService outcome is “Job status mismatch, discarding save.” Both also log completeProcessingScript success and the associated TTL message.
- The logs include 13 Hibernate INSERT statements targeting failed_jobs. Those SQL statements have no per-statement timestamps, so their execution time and transaction-commit duration cannot be measured from this capture.

Timestamp conversion audit

Sample values from the first job, 01a14a87-cfee-723a-9b35-da410e6afb4a. All events in this sample occur on 17 October 2026. Converted IST clock times are derived from the exact 13-digit epoch-millisecond values using integer millisecond arithmetic and Asia/Kolkata (+05:30); full ISO timestamps remain in the companion CSV. The logger and embedded epoch can differ: the query-start value converts to 21:12:57.097+05:30, while its log line is stamped 21:12:57.099+05:30. Both are retained as recorded.

Event	Raw epoch ms	IST time from epoch	Logger time (+05:30)	Source line
Worker query-start marker	1792251777097	21:12:57.097+05:30	21:12:57.099+05:30	Line 3
First sleep start	1792251777100	21:12:57.100+05:30	21:12:57.100+05:30	Line 5
First recoverOneJob start	1792251790339	21:13:10.339+05:30	21:13:10.339+05:30	Line 7
First recoverOneJob end	1792251790340	21:13:10.340+05:30	21:13:10.340+05:30	Line 8
Max-retries discard decision	1792251805343	21:13:25.343+05:30	21:13:25.343+05:30	Line 17
Original sleep-end marker	1792251812100	21:13:32.100+05:30	21:13:32.100+05:30	Line 43
Stuck Scheduler discarded-at marker	1792251805446	21:13:25.446+05:30	21:13:25.446+05:30	Line 42

Interpretation limits

- A timestamp marker provides millisecond resolution only. A start/end pair with a 0 ms difference means both markers fell within the same recorded millisecond; it does not prove the underlying operation took zero time.
- DB-write-start/end markers bracket the application flow, but every such attempt in this capture is followed by a status-mismatch rejection. They do not measure a successful result save or the SQL execution time.
- The Hibernate INSERT into failed_jobs and surrounding application logs prove that the INSERT was logged and that later application markers exist. Since Hibernate SQL output has no timestamp, it does not establish the exact MySQL execution time or commit completion time.
- The report describes observed log order and measured intervals only. It does not infer why recovery is triggered before the 35-second sleeps end, nor does it decide whether completeProcessingScript success changes any authoritative state beyond what is explicitly logged.

Source and companion data

Primary source: Pasted text log captured on 17 October 2026, 793 source lines. The first and last timestamped log entries are 2026-10-17T21:12:57.006+05:30 and 2026-10-17T21:24:55.649+05:30.

Companion CSV: TCS_Stuck_Job_Attempt_Timing_Detail.csv — 26 rows, one per worker attempt; includes raw epoch values, converted IST values, logger timestamps, event and source-line references, job-level discard/DLQ timestamps, and calculated intervals.